

Design of 98.8% Efficient 400-to-48-V *LLC* Converter With Optimized Matrix Transformer and Matrix Inductor

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Abstract—In recent years, there has been a considerable upswing in power usage by data centers. As a result, the 48-V power architecture has become increasingly favored due to its high efficiency. However, the cost of efficient power supplies remains challenging due to their complicated magnetic designs. The search for efficient and cost-effective power supplies continues. This article focuses on the implementation of the dc–dc stage of the power supply unit for the 48-V data center bus architecture. In this article, we present a design process for a high-efficiency high-power-density 3-kW 400-to-48-V *LLC* converter for use in data centers. Through the optimization of the matrix transformer’s geometry, turn number, and resonant frequency, we achieve superior efficiency and power density compared to state-of-the-art designs. To address challenges with the printed circuit board (PCB)-based inductors, we propose the use of a resonant matrix inductor, which reduces the number of stacked PCB layers, simplifies the winding implementation, and lowers the ac loss. By integrating the matrix inductor with the matrix transformer, we create a modular and scalable magnetic structure suitable for a wide range of voltages and currents. We also introduce electromagnetic interference shielding for the matrix transformer with a full-bridge rectifier, resulting in an 18-dB reduction in common-mode noise. The prototype achieved a peak efficiency of 98.8% and a power density of 600 W/in³, demonstrating the effectiveness of the proposed design.

Index Terms—Data center, high frequency, *LLC* converter, matrix inductor, matrix transformer, printed circuit board (PCB) magnetics, wide bandgap (WBG), 48-V bus architecture.

I. INTRODUCTION

THE growing use of cloud computing services has resulted in a significant increase in spending on data center systems, with global annual expenditure now exceeding \$200B [1]. A significant portion of this spending goes toward the physical infrastructure required to provide power, cooling, and space for server machines. According to a report from *Nature* magazine [2], the total electricity consumption of the information and communication technology (ICT) sector is projected to reach 8000 TWh by 2030, as shown in Fig. 1, accounting for around

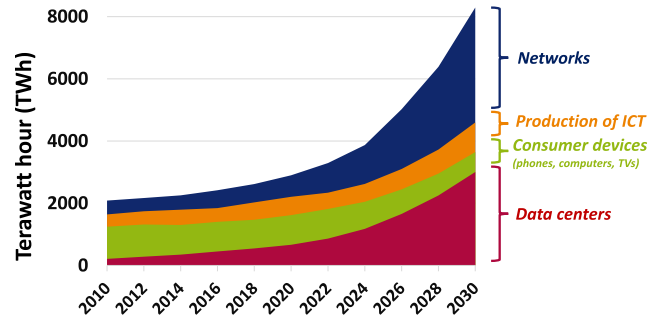


Fig. 1. Energy forecast of ICT electricity consumption [2].

20% of global electricity consumption. Data centers, in particular, are estimated to consume about 40% of the ICT sector’s energy. These trends have motivated efforts to improve the cost efficiency of data center construction, either by reducing initial construction costs or by increasing power usage effectiveness to reduce running costs.

The legacy 12-V data center bus architecture suffers from high power loss on the 12-V bus and wasted loss on the energy storage uninterruptible power supply (UPS). As a result, the 12-V bus architecture has limited power delivery capability and cannot meet the increasing power requirements of data centers. In recent years, the emergence of hyperscale data centers has made it more feasible for data centers to adopt new infrastructure, leading to a shift from the 12-V bus to the 48-V bus (see Fig. 2). The 48-V bus can deliver four times the power with the same power loss on the bus, and the battery bank unit (BBU) connected in parallel with the load eliminates UPS loss. The Open Compute Project (OCP) released the open rack V2 standard for the 48-V bus architecture in 2016 [3], specifying a nominal bus voltage of 54.5 V that should be controlled within the range of 40–60 V to follow the BBU charging profile, as shown in Fig. 2(a). This solution is similar to the existing telecom architecture and creates an ecosystem of 48-V solutions for the information technology industry. However, this solution leads to an overdesigned 400-to-48-V dc–dc stage to meet output voltage regulation requirements, limiting the expected efficiency of the power supply unit (PSU) to around 97%. In 2020, the OCP released the open rack V3 standard [4], which modified the bus architecture to a narrow-range 48-V bus with a fixed bus

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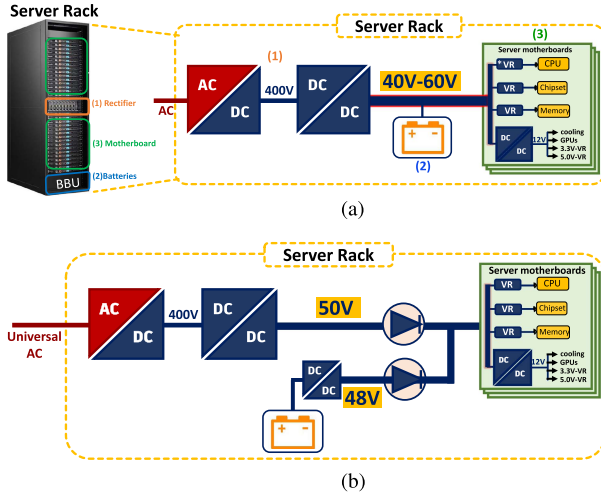


Fig. 2. 48-V bus architecture. (a) Wide-range 40–60-V bus—open rack V2 [3]. (b) Narrow-range 48-V bus—open rack V3 [4].

TABLE I

REQUIRED DC–DC STAGE EFFICIENCY ACCORDING TO THE OPEN RACK V3

	PSU [4]	PFC [5]	Required dc-dc
Peak Efficiency	97.5%	99%	98.5%
Full load Efficiency	96.5%	98.9%	97.6%
30% Load efficiency	96.5%	98.7%	97.8%

voltage of 50 V during normal power delivery conditions, as shown in Fig. 2(b). In the event of an energy shortage, power flow is naturally switched to the BBU when the bus voltage drops to 48 V. This solution offers multiple advantages, such as the following.

- 1) Narrow voltage range eliminates oversized current/voltage design.
- 2) Enabling efficient 4:1 fixed-ratio converter to downstream conventional 12-V point-of-load converters.
- 3) Power flows naturally from either PSU or BBU without dependence on software control.
- 4) Simple power supply design with fixed output voltage.

With these benefits, the narrow-range 48-V bus PSU is expected to achieve $>97.5\%$ peak efficiency and minimum efficiency of 96.5% from 30% to 100% load. Given the $>99\%$ efficiency of the power factor correction (PFC) stage [5], the required dc–dc efficiency is calculated and summarized in Table I, which indicates a required peak efficiency of $>98.5\%$ and full-load efficiency of $>97.6\%$. Some studies on server power consumption [6], [7], [8] reported that the servers nominally consume around 50% load most of the time. This is important during the power supply design to increase the peak efficiency at half-load. The 48-V power supply is also required to feature the following:

- 1) less than 500-mV pk-pk output voltage ripple;
- 2) hold-up time of 20 ms at full load;
- 3) Capable of the soft startup at full load.

The PSU is fed by a single-phase ac supply. The PFC creates double-line frequency on the intermediate 400-V bus. This ripple

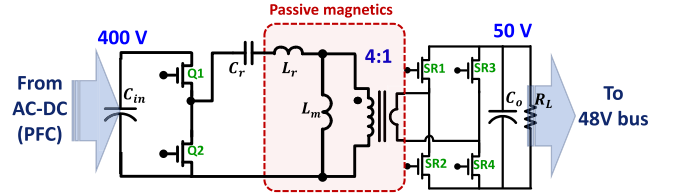


Fig. 3. HB LLC converter for 400-to-48-V 3-kW data center power supply.

is regulated through the LLC stage. Therefore, the LLC must have sufficient step-up or peak gain to regulate the PFC ripples. The LLC must also step up the voltage gain during energy shortages to meet the hold-up time requirement. In addition, the LLC must have sufficient resonant inductance to achieve soft startup and limit the inrush current with reasonable startup switching frequency [9]. Therefore, the LLC converter of the narrow-range 48-V PSU must implement a controlled resonant inductor L_r in series with the transformer (see Fig. 3) to satisfy the PSU specifications.

Utilizing wide-bandgap (WBG) technology and a soft-switching resonant LLC converter allows the switching frequency to be pushed high enough to migrate from the traditional labor-intensive Litz-wire magnetics to planar printed circuit board (PCB) magnetics. This paradigm shift provides a low-cost power supply, in which the assembly of all of the converter's components, including magnetics, is entirely automated. PCB magnetics feature multiple advantages over Litz wire.

- 1) Fully automated fabrication reduces the cost of the converter.
- 2) PCB windings feature less leakage between transformer windings because of the flexibility of the winding interleaving and reduced number of turns.
- 3) There is better thermal management due to the increased surface-to-body ratio.
- 4) Repeatable and predictable design and parasitics
- 5) The design has a low profile and high power density.

The magnetic design and integration of the LLC transformer and the resonant inductor are crucial to achieving high efficiency and power density and reducing converter cost. There have been numerous efforts to study the integrated design of passive magnetics, including inductors and transformers, as integration is known to increase power density. In [10], a hybrid Litz planar transformer is proposed where Litz wire is used for primary winding, and multiple PCB boards are stacked for the secondary winding. In [11], a similar concept is used; however, the Litz primary winding is replaced with multiple six-layer flexible PCB boards, and the secondary winding is implemented on multiple 14-layer PCB boards. Such a method can reduce the winding loss by paralleling many layers; however, the current sharing between layers is unbalanced, and the implementation is costly and complex [12]. The concept of matrix transformer was first introduced in [13], where a single transformer with a high turn number is broken down into multiple elemental transformers. Such an approach offers flexibility to the transformer design and simplifies the winding by reducing the turn number per elemental transformer. The matrix transformer with

parallel secondary has proven to provide benefits in high-output current applications [14]. In [15] and [16], the concept of flux cancellation is leveraged to integrate two transformers into a UI core for matrix transformers, where this work was applied on the 48-to-12-V bus converter [15] and 400-to-12-V front-end *LLC* converter [16]. In [17], multiple UI cores are used to increase the number of elemental transformers and further reduce the winding loss. In [18], a four-leg core and PCB winding layout were proposed to integrate four elemental transformers into one magnetic core. The three-phase interleaved *LLC* converter was also introduced to reduce the output voltage ripples and the high conduction loss of the transformer by sharing the power across the three interleaved *LLC* phases [19], [20]. However, the implementation is more complex than the half-bridge (HB) *LLC* and requires $3\times$ more primary devices. The implementation of the resonant inductor represents another challenge. The PCB-based inductor windings exhibit high ac loss because of the proximity effect and magnetomotive force (MMF) build-up. This creates high winding loss and thermal stress in the inductor winding. Besides, high stray capacitance is found between the PCB inductor layers, creating high-frequency ringing loops within the resonant circuit. The Litz-wire winding has been adopted in multiple research efforts for the resonant inductor in the 3-kW power supply dc-dc converter to avoid the high ac loss in the inductor winding [11], [21], [22]. However, this approach negates the benefits of using a PCB-based transformer as it complicates the assembly and increases the cost. In [23], an integration method for both inductors and transformers was proposed, which involves adding an extra UI core on the primary side where the primary winding serves as the inductor winding. A similar concept was used in [24] to implement a 3-kW *LLC* converter. In [25] and [26], the inductor UI core was located on the secondary side to implement a *CLL* resonant circuit. In [27], the inductance is integrated with the matrix transformer by adding a center leg between four transformer legs and rearranging the secondary windings to implement a *CLL* resonant circuit. Such a method showed high-power-density integration but only works with a center-tap rectifier. In [28], a magnetic shunt was proposed to increase transformer leakage in the *CLL* resonant converter by inserting a magnetic shunt with higher permeability between the primary and secondary windings. In [29], the magnetic shunt is reconfigured to create resonant inductance in primary for the *LLC* circuit. While the magnetic shunt demonstrated high-power-density integration, it also added complexity to the magnetic design, particularly if the interleaved winding is desired. Some studies have attempted to eliminate the need for a resonant inductor by designing unregulated DC transformer (DCX) 48-V converters [30], [31], which must be supplied by a pure dc source such as a solid-state transformer [32]. However, this approach requires significant changes to the data center power architecture and is still being researched.

In this article, we present a design process for a high-efficiency high-power-density 3-kW 400-to-48-V *LLC* converter for use in data centers. To achieve superior efficiency and power density, we optimize multiple parameters of the integrated matrix transformer, including transformer geometry, turn number,

and resonant frequency. Our design process results in a converter that outperforms state-of-the-art designs in terms of efficiency and power density.

To address the challenges associated with PCB-based inductors, we propose the concept of a resonant matrix inductor. The matrix inductor reduces the number of stacked PCB layers, simplifies the winding implementation, and lowers ac loss by distributing winding loss laterally instead of vertically on stacked layers. It also offers lower stray capacitance and fringing loss compared to multilayer single-PCB inductors. By integrating the matrix inductor with the matrix transformer, we achieve a modular and scalable magnetic structure that can be used across a wide range of voltages and currents.

In addition, we introduce electromagnetic interference (EMI) shielding for the matrix transformer with a full-bridge (FB) rectifier and present measurements of EMI noise before and after shielding, showing an 18-dB reduction in common-mode (CM) noise.

This article provides a more in-depth analysis of the design and optimization process of the *LLC* matrix transformer, building upon the ideas presented in a previously published conference paper [33]. The rest of this article is structured as follows. In Section II, we describe the structure of the proposed converter, including device selection, resonant tank design, and transformer structure. In Section III, we present the optimization of the matrix transformer's geometry, turn number, and resonant frequency. In Section IV, we introduce the concept of the resonant matrix inductor and provide simulation results demonstrating its benefits. In Section V, we describe the integration of the matrix inductor with the matrix transformer, as well as the EMI shield of the integrated magnetics. In Section VI, we present hardware results for the 3-kW *LLC* converter, offering a peak efficiency of 98.8% and power density of 600 W/in³. Finally, Section VII concludes this article.

II. DEFINING THE *LLC* CIRCUIT STRUCTURE

In this section, we define the *LLC* circuit structure, that is, the resonant tank parameters, power semiconductor devices, and transformer structure.

A. Defining Resonant Tank Parameters

The gain curve of the *LLC* is a function of the resonant tank parameters, which are the parallel-to-series resonance inductance ratio L_n and the quality factor Q , which are calculated as follows:

$$L_n = \frac{L_m}{L_r}, Q = \frac{Z_c}{n^2 R_L}, Z_c = \sqrt{\frac{L_r}{C_r}}, f_o = \frac{1}{2\pi\sqrt{L_r C_r}} \quad (1)$$

where L_m is the magnetizing inductance, L_r is the resonant inductance, C_r is the resonant capacitor, n is the transformer turns ratio, R_L is the load dc resistance, and Z_c is the characteristic impedance. The goal of the resonant tank design, including L_n and Q , is to minimize loss while still achieving the necessary maximum gain to meet the holdup requirement. According to [34] and [35], for an *LLC* running at resonant frequency ($f_o = f_s$), the *LLC* ac primary and secondary RMS

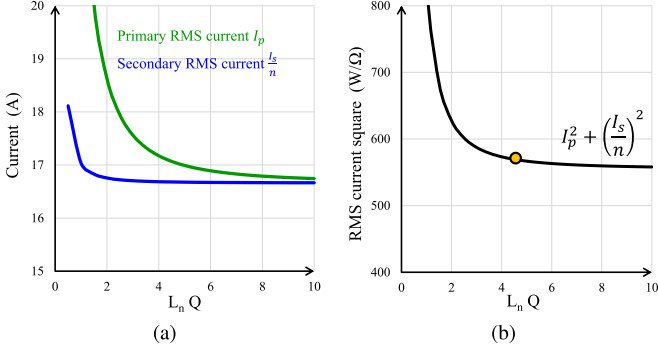


Fig. 4. Impact of $L_n Q$ on (a) primary and secondary RMS current and (b) the sum of the squares of primary and secondary RMS current (indicative of loss).

currents can be represented as

$$I_p = \frac{1}{4\sqrt{2}} \frac{V_o}{nR_L} \sqrt{\frac{n^4 R_L^2}{L_m^2 f_o^2} + 4\pi^2} \quad (2)$$

$$I_s = \frac{\sqrt{2}\sqrt{3}}{24\pi} \frac{V_o}{R_L} \sqrt{\frac{(5\pi^2 - 48)n^4 R_L^2}{L_m^2 f_o^2} + 12\pi^4}. \quad (3)$$

Equations (2) and (3) represent the RMS currents as a function in magnetizing inductance L_m and resonant frequency f_o . However, neither f_o nor L_m is readily known, and there is no clear relation between the resonant tank parameters (L_n, Q) and the LLC currents. By substituting for f_o and L_m from (1) into (2) and (3), a more generalized expression for I_p and I_s as function in L_n and Q can be derived as

$$I_p = \frac{1}{4\sqrt{2}} \frac{V_o}{nR_L} \sqrt{\frac{4\pi^2}{L_n^2 Q^2} + 4\pi^2} \quad (4)$$

$$I_s = \frac{\sqrt{2}\sqrt{3}}{24\pi} \frac{V_o}{R_L} \sqrt{(5\pi^2 - 48) \frac{4\pi^2}{L_n^2 Q^2} + 12\pi^4}. \quad (5)$$

Equations (4) and (5) are functions in L_n and Q , which are generic resonant tank characteristics and independent of the resonant frequency or the values of the passive components (L_m, L_r , or C_r). Fig. 4(a) shows the impact of the $L_n Q$ product on the LLC RMS current. The low $L_n Q$ value increases the circulating energy in the resonant tank, resulting in a high RMS current, especially on the primary side. Fig. 4(b) shows the sum of RMS current square (I^2), which is an indication of conduction losses ($I^2 R$) the LLC has to endure. The $L_n Q$ product should be maximized to reduce the RMS current; in turn, the conduction loss can also be reduced. The $L_n Q$ should be selected at the knee point of Fig. 4(b). $L_n Q$ equal to 4.3 satisfies such a requirement. Increasing $L_n Q$ more than necessary will have an adverse impact on the gain characteristics. High $L_n Q$ will move the peak gain to a much lower frequency, as shown in Fig. 5(a), making frequency modulation harder and increasing the RMS currents at high-gain operation.

The resonant tank design must also meet the gain requirement of the LLC converter. In order to maintain the output voltage during the holdup time, the converter needs a step-up gain of

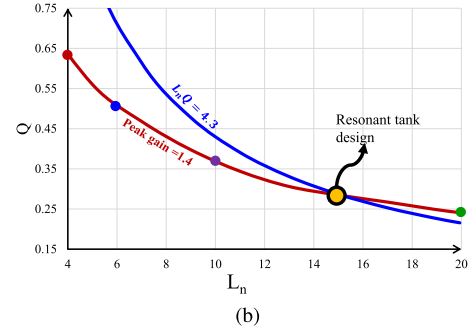
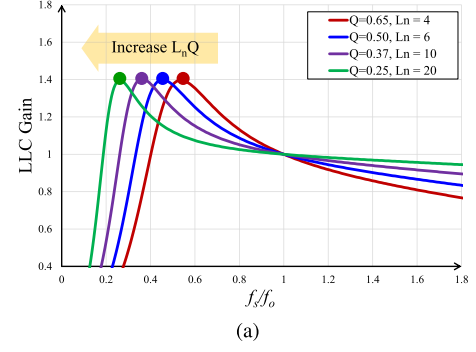


Fig. 5. L_n, Q design. (a) L_n and Q combinations to achieve a peak gain of 1.4. (b) Design point (L_n, Q) to satisfy low loss and gain requirement.

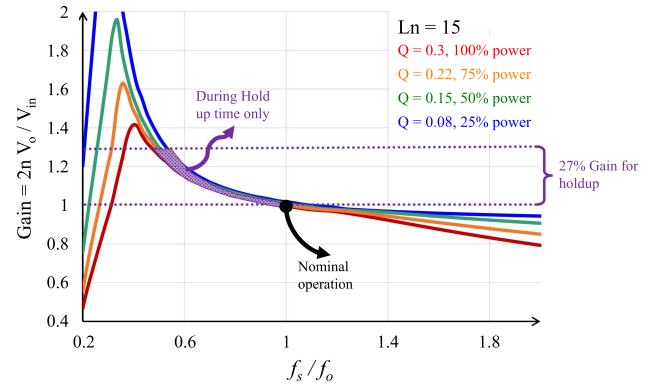
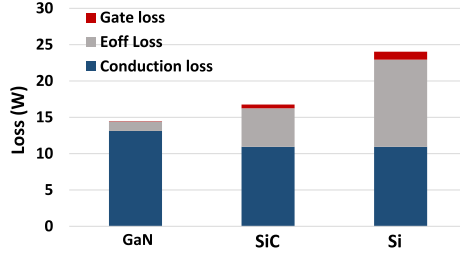


Fig. 6. Simulated gain characteristics and operation region of the LLC converter at $L_n = 15$.

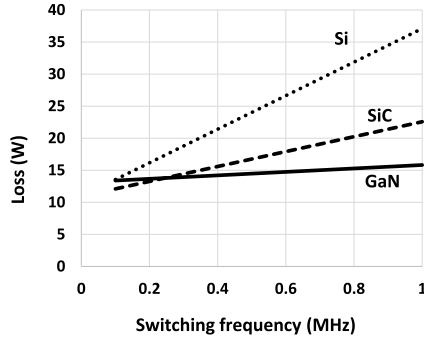
1.27. It is important to add a margin to the peak gain to avoid falling into the capacitive region, where there is a risk of losing ZVS. Fig. 5(a) shows different L_n, Q values that achieve peak gain of 1.4. The final L_n and Q values should meet both the low loss and peak gain goals. Fig. 5(b) illustrates the (blue) contour representing the low loss requirement ($L_n Q = 4.3$) and the (red) contour representing the gain requirement (peak gain = 1.4). The intersection point represents the L_n and Q values simultaneously meeting both requirements. The chosen design, with $L_n = 15$ and $Q = 0.3$, is used in the later stages of the design process and is shown to have the desired gain characteristics in Fig. 6. This design is independent of frequency or passive component values. For a given resonant frequency f_o , the passive components can

TABLE II
600-V DEVICE CANDIDATES

Technology	GaN	SiC	(5) Si
Part number	IGO60R042D	UF3SC06503	IPT60R035
C_{oss} (pF)	205	520	1604
$R_{dson-max}$ (mΩ)	42	35	35
E_{off} (μJ)*	1.3	5.3	12
Q_g (nC)	15	43	109
V_g (V)	3.3	12	10

*Simulated using LTspice at $I_D=10$ A.

(a)



(b)

Fig. 7. Evaluation of 600-V devices. (a) Loss breakdown at full load at 500 kHz. (b) Total loss at full load versus frequency.

be calculated as

$$L_m = L_n Q \frac{n^2 R_L}{2\pi f_o}, \quad L_r = \frac{L_m}{L_n}, \quad C_r = \frac{1}{(2\pi f_o)^2 L_r}. \quad (6)$$

B. Device Selection

In comparison to Si devices, WBG devices offer lower drain-source resistance, reduced gate charge (Q_g), and lower output capacitance (C_{oss}). These characteristics result in reduced switching loss, improved efficiency, and higher power density in converters. In this study, WBG devices are evaluated and compared to Si devices for use as both primary and secondary devices.

1) *Primary-Side Devices*: In the evaluation of the primary-side FETs, a quantitative comparison is made between three technologies: Si, SiC, and GaN. Table II presents three 600-V candidates for the input HB inverter. Si FETs have approximately $8\times$ higher C_{oss} and Q_g compared to the GaN FET. Fig. 7(a)

TABLE III
80-V DEVICE CANDIDATES

Technology	GaN	Si
Part number	EPC2029	BSC025N08LS5
V_g (V)	5	5
C_{oss} (pF)	820	840
$R_{dson-max}$ (mΩ)	3.2	3.2
Q_g (nC)	13	44

illustrates the loss breakdown of the 600-V devices at 500 kHz and full load. All devices are compared under the same resonant tank conditions described in Section II-A. As expected, the GaN FET exhibits the lowest loss due to its low switching loss. It is beneficial to examine the effectiveness of GaN devices across a wide range of frequencies in order to perform a comprehensive frequency optimization study on the resonant converter. Fig. 7(b) presents the loss of the GaN FET in comparison to the best Si and SiC FETs as a function of the switching frequency. The GaN FET exhibits lower loss over a wide range of frequencies, from 200 kHz to 1 MHz.

2) *Synchronous Rectifier Devices*: Similarly, the performance of 80-V GaN and Si FETs is compared for use as synchronous rectifiers (SRs). Table III shows the 80-V device candidates from GaN and Si. The GaN FET is typically driven at 5-V gate voltage. However, the Si FET can tolerate a higher gate voltage of up to 10 V. The Si performance under 5- and 10-V driving is listed in Table III. At 80 V, the gap between the GaN and Si is narrowing. Typically, Si devices can provide lower R_{dson} than GaN devices; however, Si consumes around three to six times higher gate loss. The driving voltage of the Si FET is a tradeoff between gate loss and conduction loss. Fig. 8(a) shows the loss breakdown of the FB rectifier with four parallel devices using the two candidates at 500 kHz, full load, and 25 °C. The Si FET loss heavily depends on the driving voltage, whereas 5-V driving offers less loss above 400 kHz because of the reduced gate loss. The 10-V driving offers less loss than the 5-V driving at low frequency due to the reduced conduction loss. Fig. 8(b) shows the performance of GaN and Si 80-V devices versus frequency. The GaN FET provides less loss compared to Si FET loss by reducing the switching (gate) loss, especially at high frequencies (>300 kHz). This means that GaN can offer better light-load efficiency than Si for high-frequency operation. This suggests that GaN can offer better light-load efficiency than Si for high-frequency operation, which is important for data center power supplies that typically operate at around 50% load [6], [7], [8]. Therefore, the 80-V GaN EPC2029 is used for the SR during the frequency optimization process.

The selection of the number of parallel SRs is based on both loss and thermal considerations. Fig. 9 shows the loss of the SRs and the corresponding junction temperature as a function of the number of parallel SRs. The junction temperature is calculated using the junction-to-ambient thermal resistance of 45 °C/W, as obtained from the device datasheet. From the loss and thermal perspectives, using three or four parallel devices appears to be a logical choice. Four parallel devices have a lower loss and more relaxed thermal performance, while using more

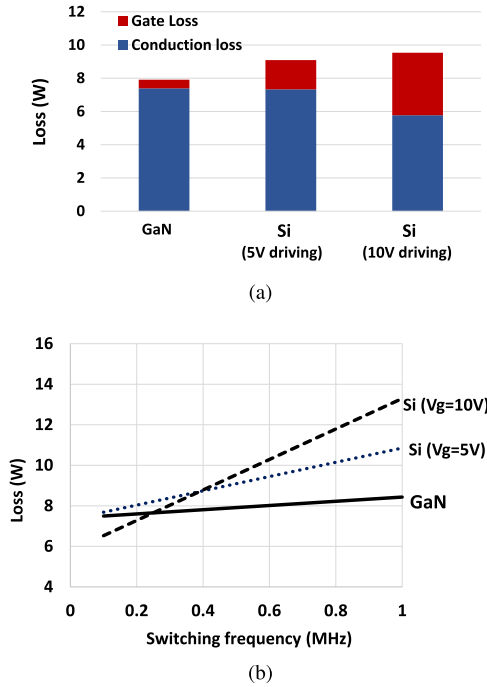


Fig. 8. Evaluation of 80-V devices. (a) Loss breakdown at 500 kHz at full load. (b) Total loss at full load versus frequency.

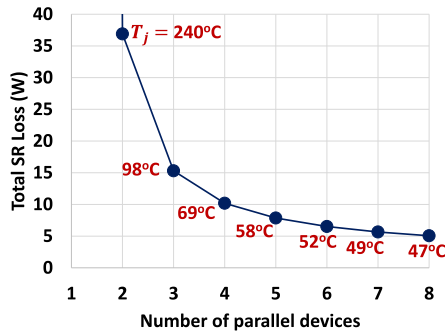


Fig. 9. Evaluation of SR loss and corresponding junction temperature (at 40 °C ambient, $R_{JA} = 45$ K/W) for GaN EPC 2029 versus the number of parallel devices.

than four devices offers diminishing returns in terms of loss and temperature reduction and is, therefore, considered to be unnecessary.

Despite the potential benefits of paralleling multiple SRs, there are also several challenges to consider.

- 1) *Current sharing between the parallel devices:* Owing to the PCB layout limitations with a high number of parallel devices, asymmetry in the power loops of the parallel devices is inevitable. Also, gate loop resistance or inductance, common source inductance, output capacitance, and ON-state resistance are other sources of unbalance. The introduction of unbalanced in-circuit parasitics between parallel devices leads to uneven sharing and degrades electrical and thermal performance, limiting the effectiveness of parallel devices [36], [37].

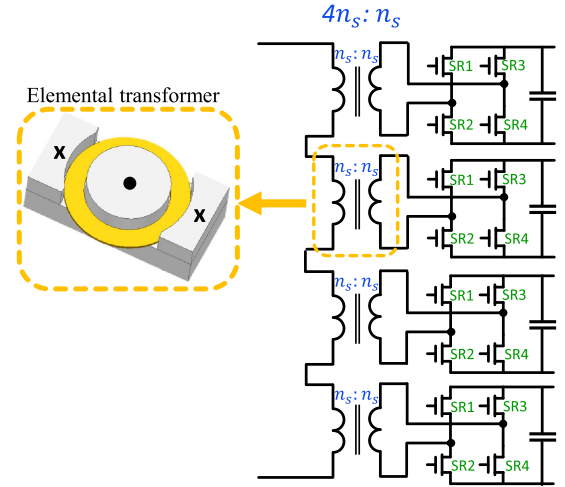


Fig. 10. Matrix of four transformers.

- 2) *High termination loss:* Paralleling devices primarily reduce conduction losses for high output currents. However, such high currents can still result in a significant loss in the PCB traces connecting the transformer's winding to the rectifier. This loss is magnified at high frequencies due to the increased ac resistance of the PCB traces.

C. Transformer Structure

To address the challenges of paralleling SRs, such as unbalanced current sharing and high termination loss, the parallel SRs can be distributed across multiple elemental transformers, as shown in Fig. 10. This four-transformer approach, also known as a matrix transformer, avoids the physical paralleling of the SRs and allows for a symmetrical power loop design and balanced current sharing. The PCB implementation of the matrix transformer also facilitates a more controlled parasitic and symmetrical construction.

Furthermore, the four-transformer approach reduces termination loss, as the termination of each FB rectifier only handles a quarter of the secondary current. It also offers a more relaxed PCB layout and improved thermal dissipation conditions.

In addition to using PCB winding, integrating the matrix transformer cores can further simplify assembly and reduce the number of magnetic components. The matrix of four transformers is integrated using the same concept as in [14], in which the elemental transformer pillars are combined under a single core plate with side legs to carry residual fluxes, as shown in Fig. 11. By using an even number of transformers and reversing the flux direction on every other transformer, the sum of the flux in the four transformer legs will equal zero (flux cancellation), and the flux in the gapped side legs will be zero. However, it is better for the flux to be distributed to the side legs in order to reduce the burden on the plate (reduce the flux density in the top and bottom plates). If the reluctances of the side legs are low enough (comparable to ferrite plate reluctance), the side legs can carry up to half of the transformer flux [24], which helps to distribute the flux more evenly in the core and reduce core loss. This can be achieved by making the side legs have zero air

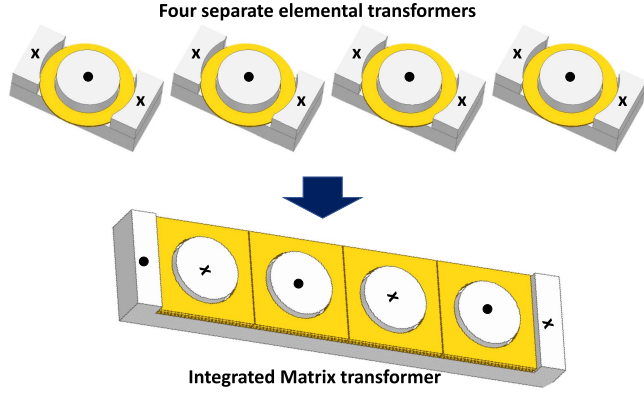


Fig. 11. Integrated matrix transformer.

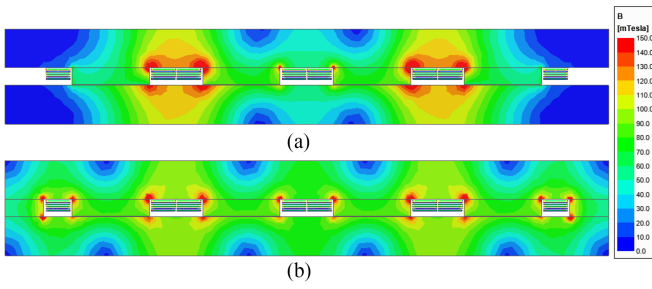


Fig. 12. 2-D FEA simulation of the flux distribution in the integrated transformer. (a) No or gapped side leg. (b) Zero airgap side leg.

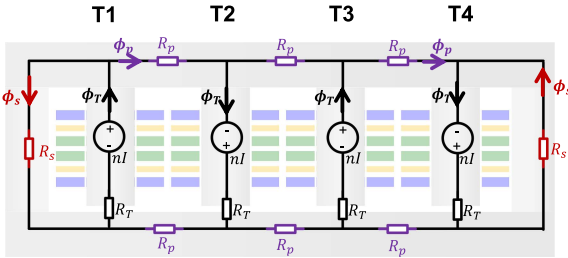


Fig. 13. Reluctance model of the integrated matrix transformer with side legs.

gap. Fig. 12 shows the flux distribution in the integrated core in the case of no side legs (or gapped side legs) and the case of zero-air-gap side legs. The zero-air-gap side legs diverge the transformer flux to the side legs, reducing the flux density in the plates (top and bottom plates). The 2-D finite-element analysis (FEA) simulation showed a 40% reduction in core losses for the simulated case shown in Fig. 12. The distribution of the flux in the side leg can be understood using the detailed reluctance model of the transformer, as shown in Fig. 13 by including the plate reluctance R_p and the side leg reluctance R_s . ϕ_s is the side leg flux, ϕ_p is the flux flowing in the plate from T1 to T2 and T3 to T4, ϕ_T is the transformer leg flux, and R_T is the transformer leg reluctance (gapped leg). Solving the reluctance model results in the following:

$$\frac{\phi_s}{\phi_p} = \frac{R_p}{R_s}. \quad (7)$$

The side leg flux is inversely proportional to the side leg reluctance R_s . For the best flux distribution ($\phi_s = \phi_p$), the side leg reluctance R_s should be close to the plate reluctance R_p . This can be realized by making the side leg have zero air gap.

III. MATRIX TRANSFORMER DESIGN

The *LLC* transformer loss represents a sizable portion of the converter loss and occupies more than 50% of the physical space of the *LLC* converter. As such, optimizing the *LLC* transformer is crucial for achieving an efficient and compact converter. The loss of the *LLC* transformer can be divided into two categories.

- 1) *Fixed loss* P_{fixed} : This loss is independent of the delivered power and remains constant at all load points. The design of the *LLC* converter is based on nominal operation at resonant frequency ($f_s = f_o$), resulting in a constant switching frequency at all load points. As a result, the transformer's switching frequency, output voltage, and volt-seconds are all unchanged with the load. This means that the core loss is independent of the output power, and the magnetizing current, which is a function of the volt-seconds, is also independent of the output power. The magnetizing current causes a circulating energy loss within the resonant tank. Therefore, the fixed losses of the *LLC* transformer include both core loss and magnetizing current loss.
- 2) *Conduction loss* P_{cond} : This loss is dependent on the delivered output power and includes the conduction loss of the transformer ($I^2 R_{ac}$). Given that the transformer RMS current is proportional to the output power, P_{cond} can be deduced to be

$$P_{cond} = I^2 R_{ac} = k P_o^2. \quad (8)$$

The ratio between the fixed loss and conduction loss determines the shape of the transformer efficiency curve. The transformer efficiency is calculated as follows:

$$\eta_{Tr} = \frac{P_o}{P_o + k P_o^2 + P_{fixed}}. \quad (9)$$

To realize the peak efficiency point, the derivative of the efficiency is equalized to zero, as follows:

$$\frac{d}{dP_o} \eta_{Tr}(P_o) = \frac{P_{fixed} - k P_o^2}{(P_o + k P_o^2 + P_{fixed})^2} = 0. \quad (10)$$

According to (10), the transformer reaches peak efficiency when the conduction loss is equal to the fixed loss ($P_{cond} = P_{fixed}$). If the circulating energy loss is relatively small compared to the core loss, we can also say that peak efficiency is achieved when the core loss is equal to the winding loss. Fig. 14 illustrates a set of pseudo-efficiency curves with 99% peak efficiency and different conduction-to-fixed-loss ratios. Design (1) has a full-load conduction loss lower than the fixed loss, resulting in low light-load efficiency. Design (3) has a full-load conduction loss 16 times the fixed loss, leading to low full-load efficiency. Designs (1) and (3) have poor overall average efficiency. In contrast, Design (2) achieves the best average efficiency by designing the peak efficiency point around the half-load. Therefore, a rule of thumb could be concluded that the transformer's

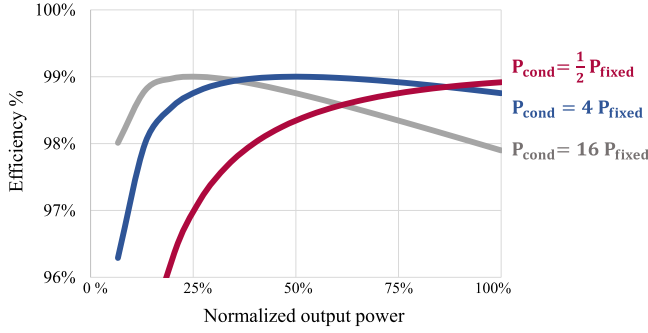


Fig. 14. Set of transformer efficiency curves with 99% peak efficiency and different conduction-to-fixed-loss ratios.

full-load conduction loss should be around $4\times$ the transformer's fixed loss, or the winding loss should equal the core loss at half load. The same rule applies to the whole converter by including the devices' conduction loss + termination loss into P_{cond} and including the device's switching loss into the fixed loss P_{fixed} . Since the design of the *LLC* focuses on the nominal operation where the *LLC* is running at resonant frequency ($f_s = f_o$), the switching frequency should be fixed at all load points. Also, when the *LLC* runs at the resonant frequency, the turn-OFF current of the primary devices equals the peak magnetizing current and is independent of the load. Therefore, at nominal operation, the device switching loss is considered independent of the load (i.e., fixed loss).

In order to achieve an efficient transformer, it is important to consider the tradeoff between fixed loss (core loss) and conduction loss (winding loss). This tradeoff can be influenced by optimizing various parameters, including:

- 1) transformer geometry;
- 2) transformer turn number;
- 3) nominal switching frequency or resonant frequency.

Optimizing these parameters is essential for designing an efficient matrix transformer and will be discussed in detail in this study.

A. Transformer Geometry

Optimizing the winding width and core cross-sectional area is a necessary step in the design of a PCB transformer. Given a fixed footprint, there exists an optimal combination of winding width (c) and core area (A_e) that minimizes loss. Previous research [17], [18] has presented a methodology for optimizing c and A_e for both rectangular and round core shapes, which involves visually inspecting loss contours through the sweeping of a range of c and D values and calculating the corresponding transformer loss, as shown in Fig. 15(a). This optimization technique enables the determination of the optimal c and D values for a given footprint. Alternatively, automated optimization programs can also be used to achieve the same results with reduced calculation effort. Fig. 15(b) presents an example of an automated Python program (`optimize_geometry`) that obtains the optimal c and D values that minimize loss for a given footprint, switching frequency (f_s), and transformer turn number (n_s).

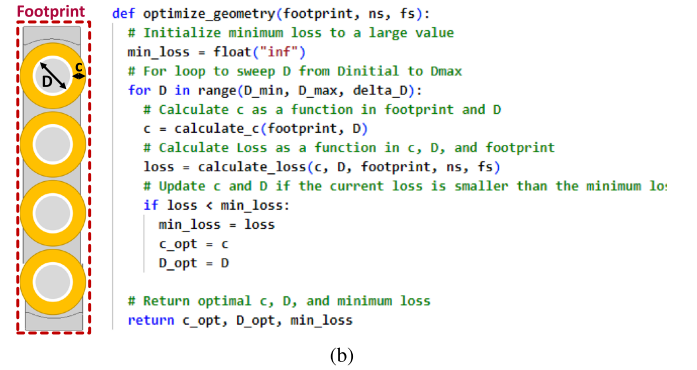
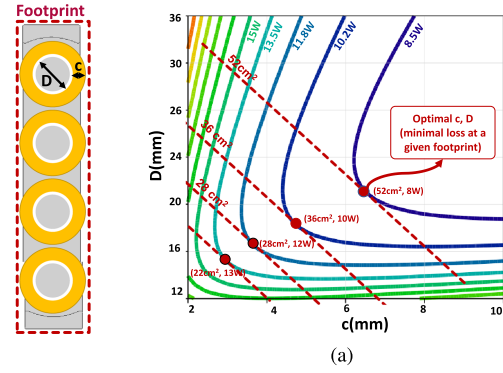


Fig. 15. Transformer geometry optimization using (a) visual inspection of loss contours and (b) using automated Python program.

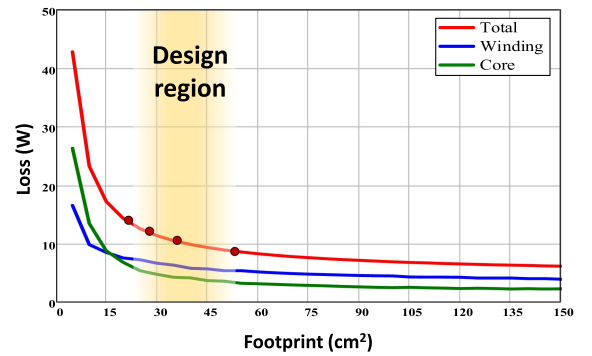


Fig. 16. Impact of the transformer footprint on the loss.

To maintain a high average efficiency for the transformer, it is recommended to maximize efficiency at half-load. Therefore, loss calculations and optimization in this study are carried out at half-load. The transformer geometry is optimized for a matrix of four 2:2 transformers at 300 kHz. Each elemental transformer has six layers of winding: two series primary layers, two series secondary layers, and two shield layers. The geometry optimization function (`optimize_geometry`) is repeated for a range of transformer footprints to evaluate the impact of the footprint on transformer loss, as shown in Fig. 16. There is an inverse relationship between loss and footprint; however, there is a region after which there is a diminishing return in loss reduction as the footprint increases. A good design region is located at the knee point, as shown in the highlighted “design region” in Fig. 16. The geometry optimization process (`optimize_geometry`)

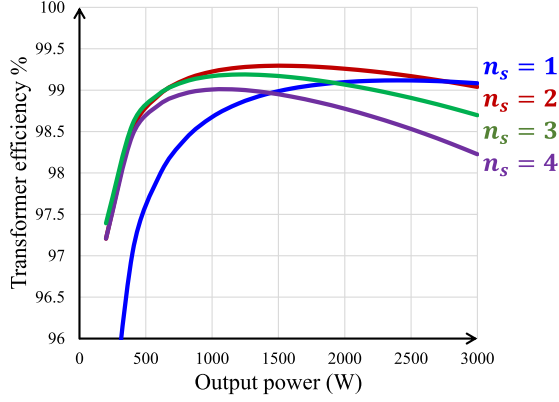


Fig. 17. Matrix transformer efficiency with different turn numbers at 35-cm² footprint.

TABLE IV
TRANSFORMER EFFICIENCY AT DIFFERENT NUMBER OF TURNS n_s [§]

turn number n_s	1	2	3	4
Average efficiency*	99%	99.2%	99%	98.9%
full-load winding loss	15.7 W	21.6 W	32 W	45 W
Core loss	11.47 W	5.1 W	4.3 W	4.2 W
Transformer volume	78 cm ³	60 cm ³	48 cm ³	42 cm ³

[§] Compared at the same footprint 35 cm² at 300 kHz

* The efficiency is averaged from 30% load to 100% load

is exercised within the next two optimization steps or whenever a design parameter, such as switching frequency f_s or transformer turn number n_s , is changed to obtain updated optimal c and D values.

B. Transformer Turn Number n_s

The transformer turn number can control the ratio between core loss and winding loss. The turns ratio of the matrix transformer is already determined ($4n_s : n_s$) to operate the *LLC* at the resonant frequency and attain high efficiency at nominal operation (400 to 50 V). This implies that the turns ratio of the elemental transformer is fixed to ($n_s : n_s$). However, the turn number n_s is still a design parameter that should be optimized. As mentioned in (11), increasing the number of turns reduces the flux density and subsequently reduces core loss. In this study, $n_s = 1, 2, 3, 4$ are considered, and the resulting losses are compared. A six-layer PCB is used, with two layers (layers 1 and 6) dedicated to the secondary winding, two layers (layers 2 and 3) dedicated to the primary winding, and two layers reserved for shielding. Fig. 17 shows the efficiency comparison between matrix transformers with different turn numbers designed at 35 cm² and 300 kHz. The transformer geometry is optimized as described in Section III-A in order to achieve the highest efficiency at the given transformer footprint. The ($n_s = 2$) case yields the highest peak efficiency. Table IV shows the averaged efficiency from 30% to 100% load for the different number of turns. The two-turn solution renders the best-averaged efficiency. The ratio between winding loss and core loss highly

depends on the number of turns. The two-turn transformer gives a winding-to-core loss ratio of approximately 4:1, which, according to the aforementioned analysis, is the best ratio. The transformer volume reduces as the turn number increases due to the smaller core resulting from the reduced B_m .

C. Nominal Switching Frequency

The *LLC* switching frequency has a significant impact not only on the transformer but also on the semiconductor loss and termination loss. This impact is depicted as follows.

- 1) The device loss increases with switching frequency as switching loss increases.
- 2) AC resistance of transformer winding increases with frequency. The rate of increase is dependent on the interleaving status of the winding [38].
- 3) The transformer core loss reduces with switching frequency if the ferrite material runs within its recommended frequency range.

The flux density in the *LLC* transformer is a function of output voltage and is expressed as

$$B_m = \frac{V_o}{4n_s f_s A_e} \quad (11)$$

where n_s is the secondary winding turn number, and A_e is the core leg cross-sectional area. By combining the Steinmetz equation ($P_v = C_m f_s^\alpha B_m^\beta$) and (11), the core loss density is expressed as a function of frequency, as follows:

$$P_v = C_m \left(\frac{1}{f_s} \right)^{\beta-\alpha} \left(\frac{V_o}{4n_s A_e} \right)^\beta \quad (12)$$

Therefore, the reduction in core loss density (P_v) due to the decrease in B_m outweighs the increase in core loss due to an increase in f_s (when $\beta > \alpha$) [15]. In general, the relationship between core loss and excitation frequency is nonmonotonic, meaning it may sometimes increase and sometimes decrease. This relationship depends on the frequency coefficient α and the flux density coefficient β . However, the values of β and α are frequency dependent. In general, α tends to increase, and β tends to decrease as the excitation frequency increases. Beyond the point where β becomes smaller than α , there is no benefit in further increasing the frequency as the core loss will increase again. For that reason, multiple ferrite materials are experimentally tested using the method in [39] at various excitation frequencies to determine the best ferrite material at each frequency. The materials are tested using small toroid samples with dimensions OD = 14 mm, ID = 7 mm, and H = 5 mm. The materials are compared based on the performance factor $B_m \times f_s$, as shown in Fig. 18. According to Faraday's law, the product $B_m \times f_s$ is proportional to the excitation voltage V_{exc} and inversely proportional to the core cross-sectional area:

$$\text{performance factor} : B_m \times f_s \propto \frac{V_{exc}}{A_e} \quad (13)$$

It is desirable to have a high-performance factor, as a material with a higher performance factor can provide more power (higher V_{exc}) or can have a smaller size at the same level of loss. The performance factor of a given material will generally

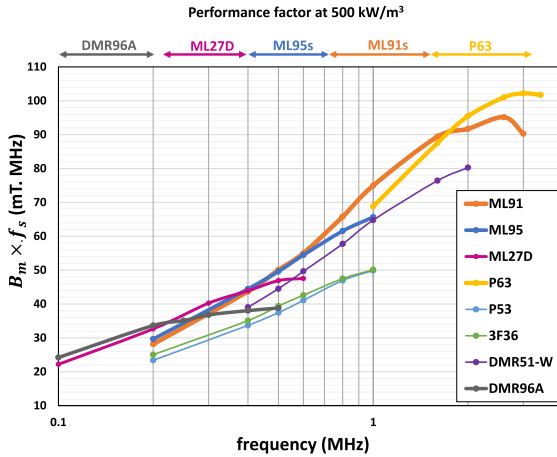


Fig. 18. Experimental evaluation of performance factor for different low-loss ferrite materials at loss density of 500 kW/m³.

increase with frequency until it reaches a peak value, after which there is no point in pushing the frequency further, and switching to a better material is recommended. Fig. 18 can be used to identify the optimal ferrite material for a given frequency by selecting the material with the highest performance factor at that frequency.

The survey presented in Fig. 18 is widely applicable and can be useful for future design projects that require low-loss materials and have similar P_v values. It is worth noting that some high-frequency materials, such as ML91s and P63, have limitations on the maximum flux density that they can sustain. For example, the manufacturer's datasheet for ML91s indicates that it should not be subjected to flux densities exceeding 50 mT to avoid material instability. In the course of optimizing the 3-kW LLC for frequency, it was found that ML95 performs better than ML91s in the 400 kHz to 1 MHz range because ML91s is limited by its maximum flux density (B_m) in this range.

The frequency optimization is a tradeoff between core loss and other converter loss components, namely, winding loss, device loss, and termination loss. The converter loss was calculated for a switching frequency range from 100 kHz to 1.6 MHz. The four-transformer matrix was optimized, and the transformer loss was evaluated at a given footprint of 30 cm². The eddy loss can be a significant part of the core loss, especially at high frequencies [24]. Since the resonant frequency optimization is explored to beyond 1 MHz, the proper estimation of the core loss is desired for fair evaluation or optimization. The core loss is evaluated according to the model in [40] and [41] to account for the eddy core loss. The transformer currents are calculated according to (4) and (5). The passive component values L_m , L_r , and C_r are calculated as in (6). The winding ac resistance is derived using the Dowell model [38] and corrected using 2-D FEA simulation for accounting for the assumptions made in the Dowell model. Fig. 19 shows the converter loss and loss breakdown. This study uses the ferrite material with the lowest core loss density for each frequency and is marked at the top of Fig. 19. It is observed that the transformer loss

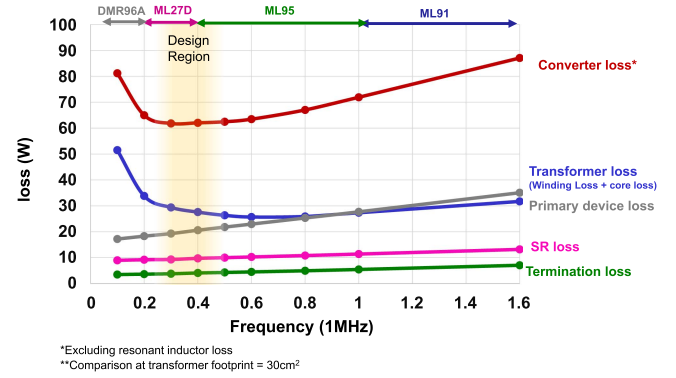


Fig. 19. Switching frequency optimization.

(winding loss plus core loss) reduces with increasing switching frequency because of the significant reduction in core loss, as described in (12). However, for frequencies above 600 kHz, the increase in winding loss surpasses the reduction in core loss, resulting in an overall increase in transformer loss, with a minimum loss occurring at around 600 kHz. The final choice of switching frequency also depends on device and termination losses, which are proportional to the switching frequency. The combined converter loss exhibits a minimum in the range of 300–400 kHz.

At this point, the transformer design has been well-defined. A matrix of four transformers is used to reduce the termination loss of the rectifier and achieve balanced current sharing among SRs. A transformer turn number of two ($n_s = 2$) is selected to attain the best efficiency. A resonant frequency of 300 kHz is chosen to minimize the converter loss. The geometry optimization discussed in Section III-A assumed a round core and round windings for simplicity in the iterative optimization process.

Finally, the transformer geometry can be further fine-tuned at the optimized parameters (footprint, turn number, and frequency) to further enhance the transformer performance. A transformer footprint of 35 cm² was deemed a reasonable choice as a tradeoff with the transformer's loss. The winding and core shapes can be elongated and optimized at this footprint. Fig. 20(a) shows the transformer dimensions, where l represents the elongation of the core, D represents the diameter of the rounded end, c is the winding width, and H is the side leg and plate thickness. This geometry is optimized as follows: for a given l value, the dimensions D and c are swept for the given footprint. The plate thickness H is calculated to make the cross-sectional area of the side legs and plates equal half of the transformer leg area ($H = 0.5A_e/(D + l)$). The loss is calculated at the swept D, c range, and the combination of D and c that gives the minimum loss is obtained. The process is repeated for a range of $l = 0 - 40$ mm, and the results are shown in Fig. 20(b). The red line shows the transformer loss at half-load versus l . Every point on the red line corresponds to optimized values of D and c . Increasing l reduces the core loss due to the improved uniformity in the flux distribution, but increases the winding loss as the mean length of the winding increases. Therefore, there is an optimal value of l , D , and c that

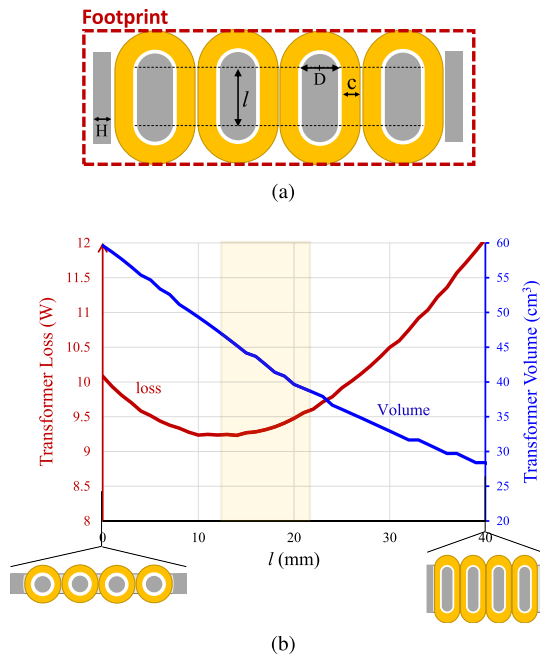


Fig. 20. Optimization of the transformer with elongated round core at 35 cm^2 . (a) Dimensions. (b) Optimization results at half-load.

minimizes the total loss. Fig. 20(a) also shows the corresponding transformer volume in blue. The elongation of the core legs reduces the transformer volume by decreasing the transformer height or the plate thickness H , which is inversely proportional to l . As a result, there is a design region (highlighted in yellow) that offers a tradeoff between volume and loss. The region to the left of the highlighted design region should be avoided, as both the loss and the volume increase. On the other hand, selecting a value of l that is larger than the optimal value can result in a smaller volume with a controlled increase in loss. The final value of l will be determined after the design of the resonant inductor, which is discussed in the next section.

IV. RESONANT MATRIX INDUCTOR

The use of PCB magnetics can reduce the cost of passive components due to automated fabrication. Integrating the magnetics into the power-stage PCB is another step toward lowering the assembly cost of the magnetic components. However, this integration can be challenging when multiple magnetic components with different characteristics are combined on the same PCB. For example, consider the case of a transformer and an inductor on the same PCB. The PCB transformer is designed with perfectly interleaved primary and secondary. As a general rule, the transformer winding layer should be designed within one to two times the skin depth. On the other hand, the inductor's winding lacks interleaving and experiences high MMF stress on the layers, making it prone to high eddy currents and eddy loss, as shown in Fig. 21(a). The PCB winding for the inductor should be designed thinner than the skin depth, in contrast to the transformer. Fig. 21(b) shows the ac resistance of an eight-layer PCB inductor versus layer thickness. This figure is based on a 2-D FEA simulation using R - Z coordinates at 300 kHz. The

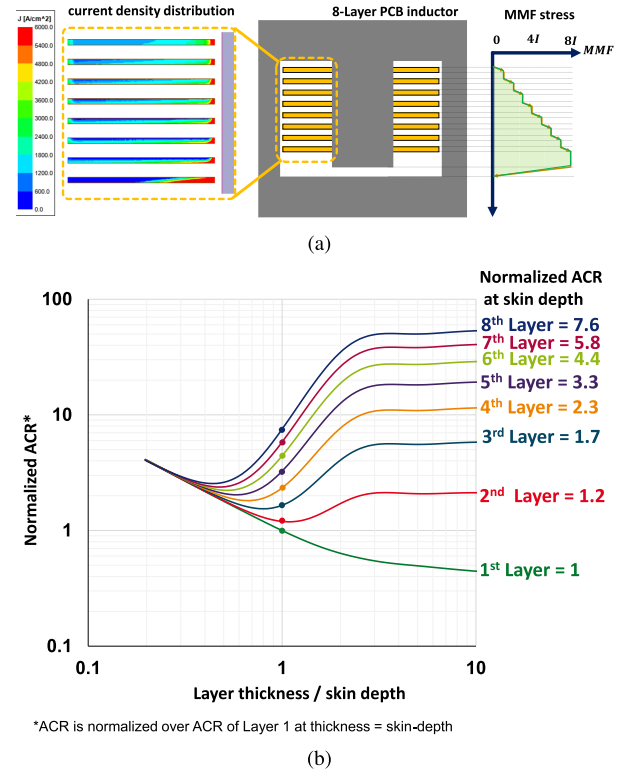


Fig. 21. Losses in eight-layer PCB inductor. (a) MMF and current density distribution. (b) AC resistance using 2-D FEA.

ac resistance is calculated by dividing the ohmic loss of the layer by the square of the RMS current and then normalized by the ac resistance of the first layer at a layer thickness equal to the skin depth. The layer thickness is swept and normalized by the skin depth (0.119 mm at 300 kHz), which is shown on the x -axis. The first inductor layer has the same MMF stress as the transformer's interleaved layers and, thus, the same ac loss factor. However, layers 2–8 have much higher ac resistance at one to two times the skin depth. For example, the eighth layer has more than seven times the ac resistance of the first layer when the thickness equals the skin depth. This adds high loss and thermal stress to the stacked inductor layers. One might argue that increasing the PCB inductor winding width will help reduce the PCB loss by reducing the dc resistance. However, in the case of the PCB inductor, the current density tends to concentrate around the inner circumference of the winding, as shown in Fig. 21(a), making it ineffective to reduce winding loss by increasing the winding width. Such challenges have led some designers [11], [21] to resort to the Litz-wire solution. However, this problem can be avoided entirely by using the matrix inductor concept.

The matrix inductor concept can be utilized in this case to break the inductor into multiple smaller inductors with fewer winding layers and lower ac loss. The air gap of the single inductor is distributed among the elemental inductors. The inductor matrix can also be integrated for easier assembly and lower core loss (see Fig. 22). The PCB matrix inductor has fewer winding layers (resulting in lower PCB cost), a smaller profile, and lower PCB winding loss. For the case of four inductors, the

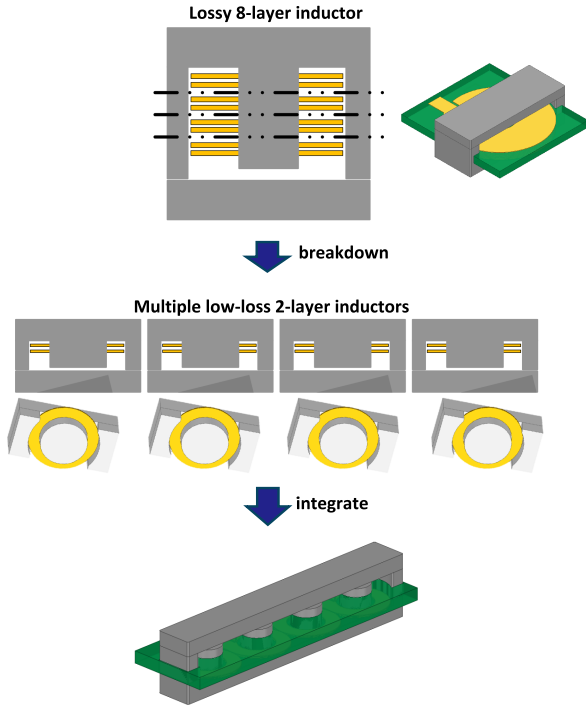


Fig. 22. Breaking down a single eight-layer inductor with loss into a matrix of two-layer inductors.

TABLE V
COMPARISON BETWEEN A SINGLE INDUCTOR AND A MATRIX INDUCTOR*

Parameter	Single inductor	Matrix inductor
number of stacked layers n_L	8	2
number of elemental inductors m	1	4
winding width c	13.2 mm	4.3 mm
Core radius r	5.5 mm	5.3 mm
Winding Loss	4 W	5.2 W
core loss	4 W	2.4 W
Turn-to-turn capacitance C_{tt}	92.8 pF	18.8 pF
Stray capacitance	13.2 pF	4.7 pF

*Compared at $L_r = 2 \mu\text{H}$ and inductor footprint = 18 cm^2 .

core volume increases by a factor of 2.5, which also increases the inductor core loss by the same factor. On the other hand, the PCB winding resistance theoretically reduces to one-third of the single inductor loss (when the core radius and winding width are fixed). This is immensely helpful for PCB inductors to reduce thermal stress on the inductor winding and convert from Litz-wire inductor. The benefits of the matrix inductor are summarized as follows.

A. Lower Winding Loss and Fringing Loss

It is clear from the previous discussion that the matrix inductor helps to reduce the MMF stress on the winding layers by distributing the inductor's turns laterally on multiple core legs. However, this also increases inductor size and core loss. To make a fair comparison, the single and matrix inductors were optimized at 300 kHz, and their losses are compared at an inductor footprint of 18 cm^2 . The loss comparison between the single and matrix inductors is shown in Table V. The matrix

inductor has lower overall inductor loss due to lower winding loss.

In addition, the matrix inductor distributes the inductor air gap across multiple elemental inductor legs. This means that a matrix of four inductors has an air gap that is one-fourth of the air gap of the single inductor designed with the same leg cross-sectional area. Distributing the air gap can help reduce the fringing effect [42].

B. Lower Stray Capacitance

The stray capacitance of the resonant inductor leads to high-frequency ringing occurring on the resonant current. This leads to higher winding loss and EMI issues. The matrix inductor reduces the winding loss by reducing the ac loss rather than reducing the dc resistance (increasing winding width), as in the single inductor case. This means that the matrix inductor uses smaller winding width and accordingly a smaller winding area. Thus, the turn-to-turn capacitance is smaller in the matrix inductor. The stray capacitor is compared for the single and matrix inductors according to [43] and modified for the matrix inductor as

$$C_{\text{stray}} = C_{tt} \times \frac{1}{n_L - 1} \times \frac{1}{m} \quad (14)$$

where C_{tt} is the turn-to-turn capacitance, n_L is the number of stacked turns, and m is the number of elemental inductors. Table V shows the calculated stray capacitance at an 18-cm^2 footprint using an eight-layer PCB of 2.8-mm thickness with FR4 dielectric. The matrix inductor capacitance is reduced to 35% of the single inductor capacitance. This is calculated assuming the matrix inductor layers are distributed on the first two layers of the eight-layer PCB, which is the worst-case scenario.

C. Less Thermal Stress

As discussed earlier, stacking inductor layers results in a huge concentration of loss on the bottom layers (layers close to the air gap). This adds high thermal stress to the PCB inductor. On the other hand, the matrix converter reduces the number of stacked layers in order to reduce the loss per layer value, which simplifies thermal management. Thermal simulations are presented here to verify the effectiveness of the matrix inductor. The single inductor is compared to the matrix inductor when both inductors were optimized at an 18-cm^2 footprint. The winding loss per turn is calculated using 2-D FEA simulations, which are then inserted into a 3-D ICEpak model. Both the inductors are simulated with forced air cooling at a flow velocity of 2 m/s (400 LFM). The single inductor [see Fig. 23(a)] has a 50°C temperature rise but the matrix inductor temperature rise is 30°C [see Fig. 23(b)]. The matrix inductor reduces the temperature rise by 40%.

Given the multiple benefits of the matrix inductor, it is adopted here to implement the resonant inductor. A matrix of four elemental inductors is chosen to match the number of elemental transformers for ease of integration, as will be shown in the next section. The matrix inductor is optimized in a similar manner as in Section III-A. Different numbers of turns or layers are explored to find the optimal number of turns for a low-loss

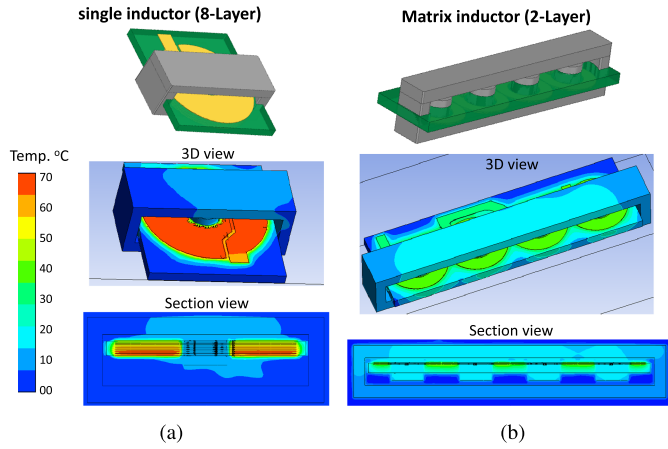


Fig. 23. Thermal simulation results at full load. (a) Single eight-layer inductor. (b) Matrix inductor on two layers.

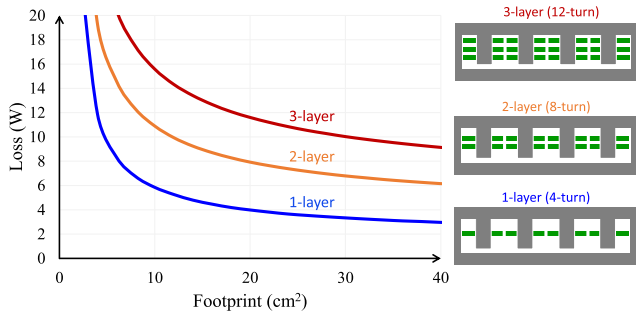


Fig. 24. Matrix inductor loss versus footprint at different numbers of turns or layers.

inductor. Fig. 24 shows the matrix inductor loss for one-, two-, and three-layer inductors. The one-layer inductor (four turns) is determined to be the better choice due to its lower loss.

V. TRANSFORMER-INDUCTOR INTEGRATION AND PCB LAYOUT

The matrix inductor and the matrix transformer are designed on the same six-layer PCB with the power devices and drivers. The matrix transformer is designed at 35-cm² footprint, and the inductor is designed at 18 cm². The transformer design point ($l = 20$ mm, $D = 9.2$ mm, $c = 4.8$ mm) is selected from Fig. 20(b) to make the matrix transformer and inductor of uniform width, as shown in Fig. 25(a). Separately laying out the inductor and transformer windings leads to two problems: 1) an increase in conduction loss in the added series interconnection between the inductor and the transformer and 2) an increased magnetic footprint to provide space for the interconnection layout. Integrating the inductor and transformer windings and cores, as shown in Fig. 25(b), can reduce the magnetic loss and footprint.

Integrating the transformer and inductor windings can help to reduce the conduction loss. The primary winding turn can wind both the transformer leg and the core leg simultaneously. This eliminates the need for an external series connection between the transformer and inductor terminals. Fig. 26 shows the 3-D FEA simulation before and after winding integration. The 3-D

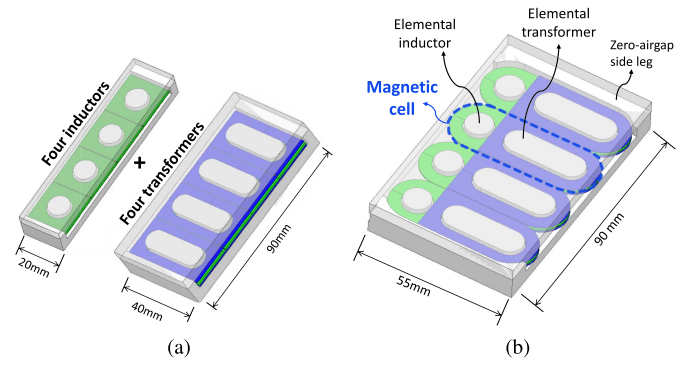


Fig. 25. Integration of matrix transformer and inductor. (a) Before integration. (b) After integration.

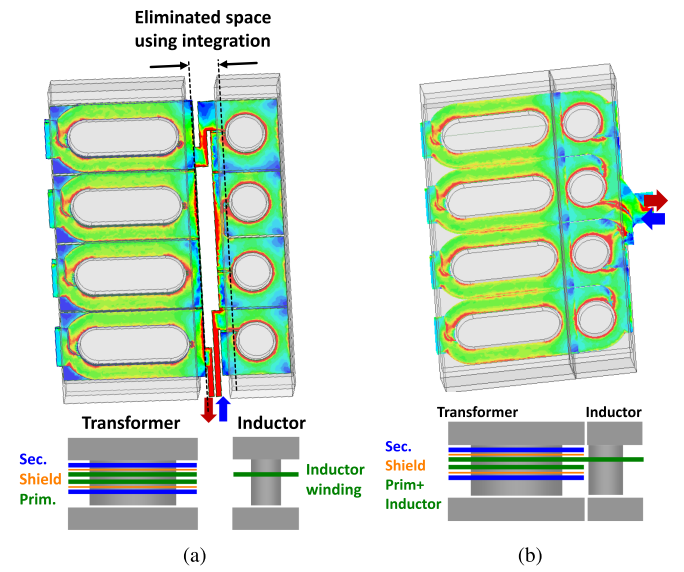


Fig. 26. 3-D FEA simulation of (a) separate transformer and inductor windings and (b) integrated transformer and inductor winding.

FEA showed a 15% reduction in winding loss (transformer and inductor combined) due to the elimination of interconnection loss and reduced overall winding length. In addition, the footprint of the integrated version is smaller after eliminating the space for the interconnection and merging the inductor winding with the primary transformer winding. A complete PCB layout is shown in Fig. 29. Furthermore, the magnetic cores of the matrix inductor and transformer can be integrated by combining all the core legs of the inductor and the transformer under one plate. This allows for a better utilization of the plate, as it allows both the inductor and transformer fluxes to pass through the same cross-sectional area of the plate. Owing to the phase shift between the two flux components, as shown in Fig. 27(a), the peak flux of the inductor does not coincide with the peak transformer flux (magnetizing flux), allowing the whole plate to be utilized for the inductor flux at time t_1 . Fig. 27(b) shows the 3-D FEA simulation of the magnetic core before and after integration when the inductor is at its peak (at t_1) at full load. The integration reduces the flux density in the plate by allowing for

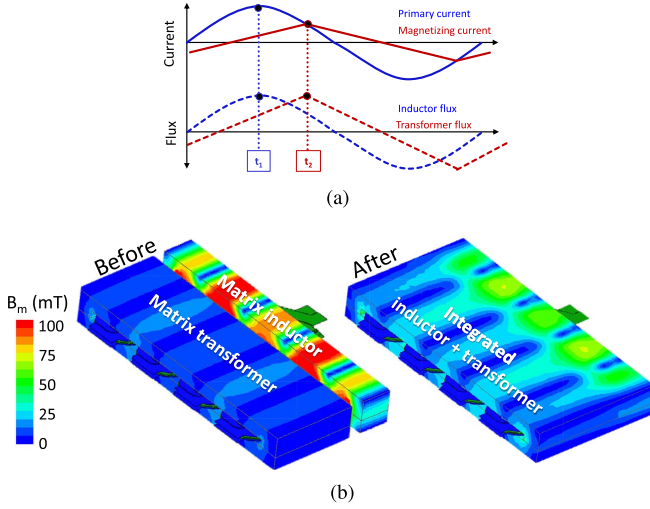


Fig. 27. Core integration. (a) Phase shift between the transformer and inductor fluxes. (b) 3-D FEA simulation of the magnetic core before integration (left) and after integration (right) at instant t_1 .

the better utilization of the plate. The same benefit is achieved during the peak transformer flux (at t_2). The 3-D FEA simulation of the core showed a 20% reduction in core loss after integration.

One elemental transformer and inductor represent a cell for the integrated magnetics. Here, four cells [as shown in Fig. 30(b)] were used as a requirement for the number of parallel rectifiers (four rectifiers). However, the integrated matrix transformer with a matrix inductor is modular and scalable to any number of elemental transformers and inductors. This scalability allows the magnetics to adapt to any number of output rectifiers and a wide range of voltage and current levels. Low-voltage high-current applications would require more magnetic cells (more elemental transformers and inductors). High-voltage low-current applications would require fewer magnetic cells. The number of magnetic cells depends on the conduction loss, volt-second, and available power semiconductor devices. The magnetizing inductance and resonant inductance of the integrated matrix transformer and matrix inductor can be calculated as

$$L_m = \mu_o n_p^2 \frac{A_{eT}}{m_T l_{gT}} \quad (15)$$

$$L_r = \mu_o n_L^2 \frac{A_{eL}}{m_L l_{gL}} \quad (16)$$

where n_p is the primary winding turn number, A_{eT} is the elemental transformer cross-sectional area, l_{gT} is the elemental transformer air gap, m_T is the number of elemental transformers, n_L is the total number of inductor turns, A_{eL} is the elemental inductor cross-sectional area, l_{gL} is the elemental inductor air gap, and m_L is the number of elemental inductors ($m_L = m_T$). The difference between the transformer and inductor air gap should be controlled during the core fabrication process by making the transformer leg longer or shorter than the inductor leg. However, for easier manufacturability, a uniform air gap might be desired. The condition for uniform air gap design is

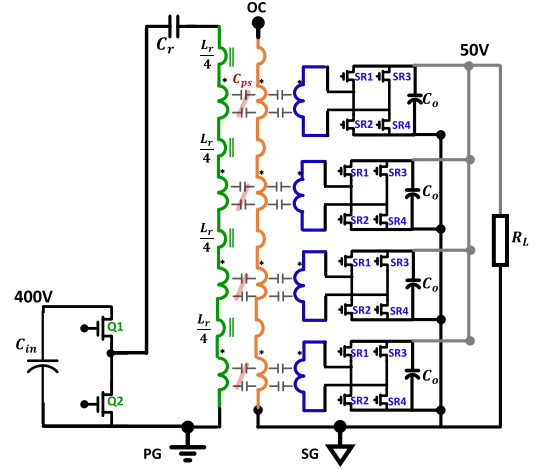


Fig. 28. Circuit diagram of the proposed integrated matrix transformer and matrix inductor showing the EMI shielding.

derived by dividing L_m/L_r as follows:

$$\frac{A_{eT}}{A_{eL}} = L_n \frac{n_L^2}{n_p^2}. \quad (17)$$

In our design, this condition is satisfied with $A_{eT}/A_{eL} = 3.75$, and a uniform air gap of 0.17 mm is used.

Fig. 28 shows the equivalent circuit diagram for the proposed integrated matrix transformer with the matrix inductor. The resonant inductance is distributed between the elemental transformers. In order to block the CM current in the primary from reaching the secondary side, the EMI shielding winding is designed [44], [45]. The shielding of a matrix transformer has been studied in [46] for a matrix transformer with a center-tap rectifier by making the shield similar to the secondary winding and connecting it to the input ground. However, when an FB rectifier is used, the secondary winding is not connected to a static-electric-potential (SEP) point, unlike the case in the center-tap rectifier. This makes it difficult to adopt the concept in [46]. In our design, the shield winding is designed to have the same shape and number of turns as the primary winding. Consequently, the differential voltage between the shield winding's terminals v_{shield} is the same as the differential voltage on the primary winding v_{primary} . The primary winding is tied to an SEP point on the input side (input ground PG). Similarly, the shield is tied to an SEP point on the secondary side (SG). In this way, the capacitance between the primary and shield layers C_{ps} is excited with static voltage, and the CM current ($I_{CM} = C_{ps} \frac{d}{dt}(v_{\text{primary}} - v_{\text{shield}}) = 0$) is blocked. This method is effective in reducing the EMI noise in FB rectifier circuits.

The final PCB layout is introduced in Fig. 29. The matrix transformer turn number is 8:2, with each elemental transformer being 2:2. The matrix inductor has four turns and is designed at $L_r = 2 \mu\text{H}$, $L_n = 15$. The shield winding takes the same shape as the primary windings, as shown in Fig. 29(c). There are no secondary layers located at the inductor legs; therefore, the thickness of the shield turns at the inductor legs is reduced to limit the eddy loss in the shield. In this design, the air gap

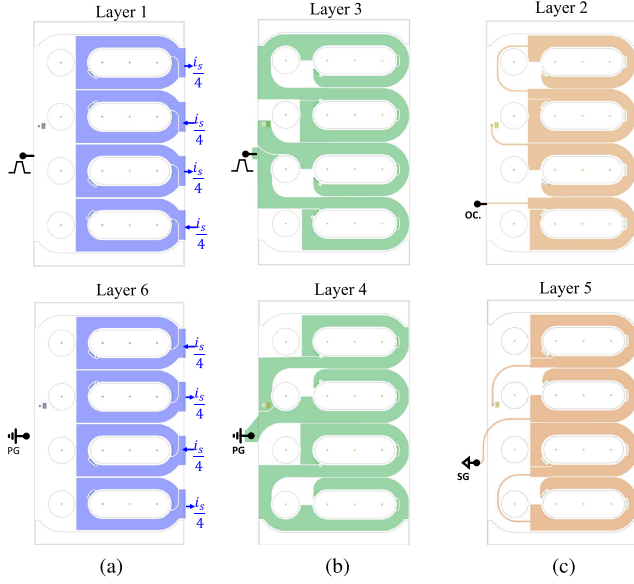


Fig. 29. PCB winding layout. (a) Secondary. (b) Primary. (c) Shielding.

length of both the inductor and transformer legs is 0.17 mm. The fringing effect is avoided by extending the core leg to space the air gap away from the PCB board [23], [42]. The ratio between the air gap spacing to the air gap length depends on multiple factors like skin depth, copper thickness, and air gap length. This ratio is obtained using 2-D FEA simulation by sweeping the core leg extension and calculating the loss. The air gap was spaced to six times the air gap length, which resulted in only a 7% increase in winding loss from fringing compared to the case with no air gap at all.

VI. EXPERIMENTAL RESULTS

The design concept is verified on a 3-kW 400-to-48-V *LLC* converter, shown in Fig. 30. The integrating magnetics, power switches, and drive circuits are populated on the same six-layer PCB. Four layers are utilized for the transformer windings and two layers for the shielding. 600-V GaN devices are used for the primary HB, and 80-V GaN devices are used for the secondary SRs. The converter is tested at $f_s = 300$ kHz, $V_{in} = 400$ V, and $V_o = 50$ V. The prototype information are listed in Table VI. The inductor and transformer core legs are fabricated with the same length since they have the same air gap of 0.17 mm. The side legs have zero air gap and are fabricated 0.17 mm longer than the transformer legs. The core is implemented with material DMR96A because of the fabrication limitations on material ML27D at the testing time. DMR96A represented the second best candidate at 300 kHz according to Fig. 18. The converter is designed to operate nominally at 400-V input and 50-V output. In the case of energy shortage (input power is disconnected), the converter is designed to support the rated output voltage of 50 V up to 20 ms at the minimum input of 315 V. The resonant tank parameters are primarily designed to meet the holdup requirement mentioned in Section II. Furthermore, as a by-product,

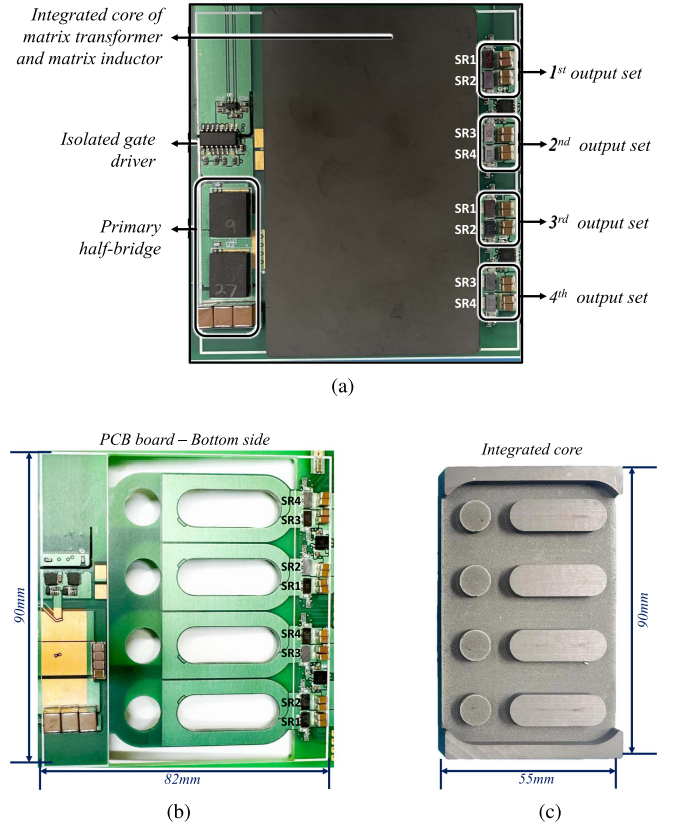


Fig. 30. Hardware prototype of 400-to-48-V 300-kHz *LLC* with integrated PCB magnetics. (a) Complete converter. (b) PCB board. (c) Core.

TABLE VI
3-kW *LLC* PROTOTYPE PARAMETERS

Parameter	Value
Primary driver	Isolated Si8273
Primary GaN device	IGT60R042D1 600 V, 42 mΩ
SR driver	HB driver LMG5113
SR GaN device	EPC2029, 80 V
Magnetizing inductance	30 μH
Resonant inductance	2.1 μH
Input voltage	400 V
Nominal output voltage	50 V
Resonant frequency	300 kHz
Printed circuit board	Six-layer, 4 oz. Cu., 62mil thick.
Core Material	DMR96A
dimensions	
Transformer	$l = 20$ mm, $D = 9.2$ mm, $c = 4.8$ mm
Inductor	$D = 9.2$ mm, $c = 4.8$ mm
Plate and side-leg	4.5 mm thickness
Converter	$3.2 \times 3.5 \times 0.46$ in

including a resonant inductor helps reduce the current stress and the switching frequency during the soft-startup process. The initial soft startup frequency that is required to keep the primary current stress within 150% of the peak nominal resonant current is calculated according to [9] and estimated to be 1.2 MHz (four

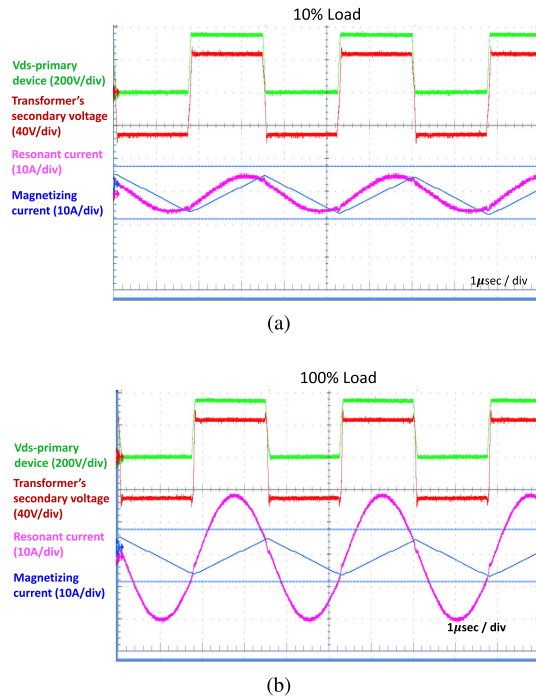


Fig. 31. Operating waveforms at nominal operation at 400-V input and 50-V output. (a) 10% load (6-A output current, 300-W power). (b) 100% load (60-A output current, 3000-W power).

times the resonant frequency). The converter has four parallel FB rectifiers distributed on the four transformers. The secondary winding has two turns distributed on the top and bottom layers. For each elemental transformer, an HB of the FB rectifier is populated on the top PCB side and connected to the secondary winding terminal on the top layer. The other HB is populated on the bottom PCB side and connected to the other secondary winding terminal on the bottom layer. The termination AC resistance (ACR) is experimentally measured according to [17] using an impedance analyzer. There are eight output capacitors on each FB, four on the top side and four on the bottom side, totaling 32 capacitors for the whole converter. Ceramic capacitors are used as they offer less parasitic equivalent series resistance (ESR) and equivalent series inductance. The output capacitor type is 1210, X7S, 100-V, 10-μF capacitor. This capacitance reduces to 2.5 μF at 50-V DC bias, and the ESR is 2 mΩ. The total installed capacitance after DC-bias derating is 80 μF. The output voltage ripple is within 300 mV. Furthermore, Ceramic capacitors 1210, C0G, 450 V, 33 nF are used for the resonant capacitor. Capacitors with a C0G temperature coefficient can offer stable operation with respect to temperature and DC bias. Fig. 31 shows the steady-state operating waveforms of the 3-kW converter at the nominal operating point ($V_{in} = 400$ V, $V_o = 50$ V). The waveforms show the drain-source voltage of the bottom switch of the input HB (green), the transformer voltage on the secondary winding (red), the primary resonant current (magenta), and the magnetizing current (blue). The magnetizing current is derived or calculated using the math function in the scope by integrating the secondary transformer voltage and

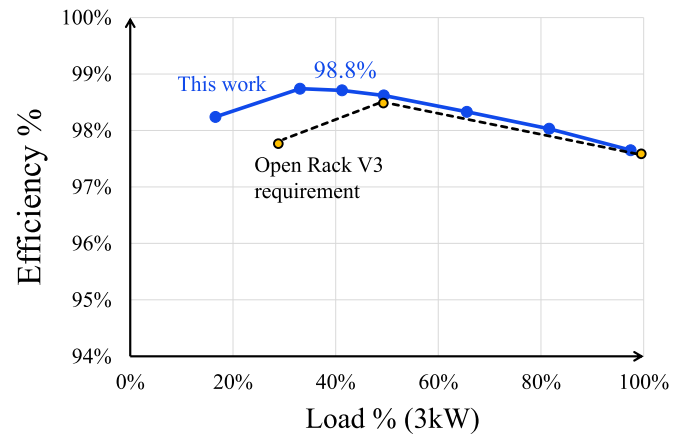


Fig. 32. Tested efficiency at 400 to 50 V.

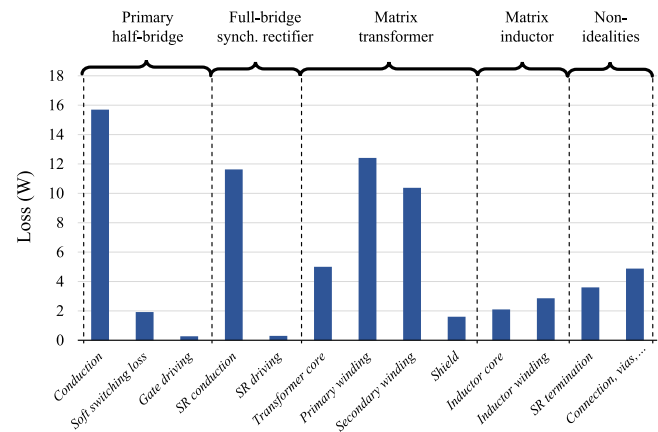


Fig. 33. Loss breakdown at full load at 3 kW.

dividing it by the magnetizing inductance. Fig. 31(a) shows the waveforms at 6-A output current and 300-W power. Fig. 31(b) shows the waveforms at 60-A output current and 3000-W power.

The converter efficiency is tested at nominal conditions using power Analyzer YOKOGAWA PZ4000 and electronic load BK PRECISION 8620. The converter can achieve 98.8% peak efficiency, as shown in Fig. 32. The peak efficiency is located at 40% load, providing high average efficiency around half-load, which is the most important for data center power supplies [6], [7], [8]. This is achieved by means of careful design of the ratio between the conduction-to-fixed loss, as shown in the loss breakdown in Fig. 33. The core loss is designed to be around one-fourth of the winding loss. This improves the light-load efficiency and shifts the peak efficiency below half-load. The termination ACR is experimentally measured according to [17] using an impedance analyzer. The shielding layers also incur conduction loss due to the induced eddy currents. The shielding loss is estimated using 2-D FEA simulation, and the losses are estimated to be 8% of the transformer winding loss. The measured efficiency excludes the gate loss. The gate drivers are supplied externally from a bench power supply. The gate driving losses are measured experimentally by calculating the

TABLE VII
SURVEY AND COMPARISON OF THE 400-TO-48-V REGULATED DC–DC CONVERTERS FOR 48-V DATA CENTER ARCHITECTURE

	[20] 2019	[26] 2020	[24] 2021	[11], [12] 2021	[27] 2022	This work
Topology	Three-phase LLC	Half-bridge CLL	Full-bridge LLC	Full-bridge LLC	Full-bridge CLL	Half-bridge LLC
Rated power	3 kW	800 W	3 kW	3.2 kW	3 kW	3 kW
Implementation	PCB magnetics	PCB magnetics	PCB magnetics	Stacked PCB Tr., Litz wire L_r	PCB magnetics	PCB magnetics
Frequency	1 MHz	700 kHz	1 MHz	550 kHz	1 MHz	300 kHz
Peak efficiency	97.7%	95.83%	98%	98.6%	97.3%	98.8%
Power density	600 W/in ³	220 W/in ³	700 W/in ³	73 W/in ³ *	550 W/in ³	600 W/in ³

*Calculated for the whole PSU (PFC + LLC).

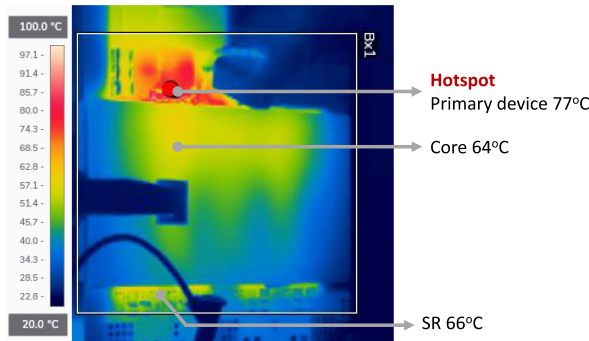


Fig. 34. Thermal performance of the 3-kW LLC converter with forced air cooling of 800 LFM.

output power of the drivers' supply. The primary driving loss is 0.27 W, and the secondary driving loss is 0.3 W. Since GaN devices are used on both primary and secondary sides, the driving losses are insignificant and combined are less than 0.6 W. GaN technology helps to maintain high efficiency at light load and idle mode by leveraging the low driving and switching losses.

This work is compared to the state-of-the-art achievements, as shown in Table VII. The proposed 3-kW *LLC* converter shows superiority in terms of implementation, efficiency, and power density.

Fig. 34 shows the steady-state thermal performance of the 3-kW prototype in the case of full power, and 800-LFM forced air cooling. The hotspot appears at the primary HB of 77 °C. The core temperature and the rectifier temperature are 64 °C and 66 °C, respectively. The manageable transformer's and rectifier's temperature is due to the designed matrix transformer with four output sets, a low profile, and a high surface-to-body ratio. The matrix transformer reduces the current stress for each output set and allows for better thermal dissipation.

The conducted EMI noise is evaluated for this converter with the proposed shielding (see Fig. 35). The EMI noise is reduced by 16 dB at the fundamental frequency, and this reduction is effective up to 20 MHz.

VII. CONCLUSION

In this article, we presented a high-efficiency low-profile design for a 48-V *LLC* converter intended for use in data centers. The converter employs PCB magnetics, which offers improved thermal management, ease of assembly, and cost-effectiveness.

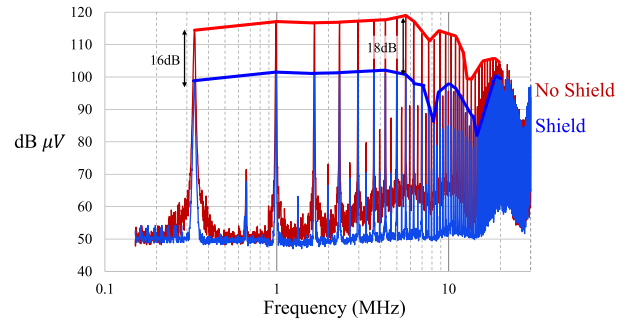


Fig. 35. Tested EMI.

However, there are several key challenges associated with PCB magnetics, including high ac loss in the transformer, high thermal stress in the PCB, particularly in PCB-based inductors, and EMI caused by high interwinding capacitance. To address these challenges, we optimized the matrix transformer through various techniques, including transformer geometry, turn number, and operating frequency. We also introduced the matrix inductor concept, which reduces PCB inductor winding loss and thermal stress on the PCB and prevents extreme increases in ac resistance due to MMF build-up through the lateral stacking of layers. The converter was integrated into a six-layer PCB with two shielding layers to block EMI noise. Our design achieved an impressive efficiency of 98.8% and a power density of 600 W/in³ with a low profile of approximately 11 mm. The low profile reduced PCB-to-air thermal resistance and facilitated thermal management. This article demonstrated the potential of the proposed converter design to meet the demanding requirements of data center power systems.

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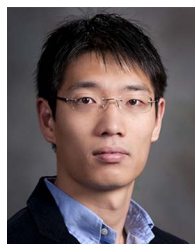
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